Vision-Mate: A Real-Time Obstacle Detection and Distance-Estimation System for Visually Impaired Users

Mohammed Asadullah Shareef

AI/ML Research Internship, International Institute of Information Technology, Hyderabad (IIITH), Oct–Dec 2025

Abstract

Vision-Mate is a real-time computer vision pipeline that fuses object detection and monocular depth estimation to provide obstacle-awareness audio alerts for visually impaired users. The system combines YOLOv8 for obstacle detection with Depth Anything V2 for per-pixel depth estimation, converting relative depth into approximate physical distance to trigger alerts when an obstacle breaches a 1.5-meter safety threshold. This paper describes the system's architecture, the video-processing pipeline built to validate it, and the accuracy metrics used for evaluation. It also documents a safety-relevant failure mode discovered during development: the system's distance estimates depended on a hardcoded calibration constant, making distance thresholds unreliable across cameras with different focal lengths. We treat this not as an implementation bug to be quietly patched, but as a case study in how assistive and safety-critical AI systems can silently fail in ways that standard accuracy metrics do not surface, and outline the calibration approach needed to generalize the system safely.

1. Motivation

Assistive navigation tools for visually impaired users must convert visual information into a form that can be acted on immediately and safely, typically audio or haptic feedback. Object detection alone is insufficient: knowing that an obstacle exists is less useful than knowing how far away it is and whether it demands immediate action. Vision-Mate was built to close that gap by combining detection with depth estimation in a single real-time pipeline, and to surface, through direct engineering experience, where such a system's guarantees break down.

2. System Architecture

The pipeline processes a live or recorded video feed through four stages:

2.1 Frame Extraction

FFmpeg extracts frames from the incoming video feed at a fixed rate, decoupling the detection/depth models from the raw stream's encoding and framerate.

2.2 Obstacle Detection

YOLOv8 performs object detection on each extracted frame, producing bounding boxes and class labels for obstacles in the scene.

2.3 Depth Estimation

Depth Anything V2 generates a per-pixel relative depth map for the same frame. The depth value at each detected bounding box's centroid is sampled and converted into an approximate physical distance using a calibration constant.

2.4 Alert Generation

Detections are color-coded by proximity, and when the estimated distance to an obstacle falls below 1.5 meters, the system triggers a temporal audio alert. Combining a visual (bounding box) and audio (alert) channel was intended to support both partially and fully visually impaired users, and to make the system's internal state inspectable during testing.

3. Evaluation

Detection accuracy was validated on a custom dataset using Precision, Recall, and F1-score. These metrics assess how reliably the system identifies obstacles present in a scene, but — as discussed below — they do not capture whether the distance estimate attached to a correctly detected obstacle is itself trustworthy.

4. Failure Mode: The Hardcoded Calibration Constant

During development, we identified that the conversion from Depth Anything V2's relative depth output to a physical distance in meters relied on a single hardcoded calibration constant, tuned to one camera's field of view and focal length. This meant the 1.5-meter alert threshold was only meaningful for the specific camera used during development. On a camera with a different focal length, the same relative depth value would correspond to a different real-world distance, silently shifting the effective alert threshold without any change in the system's detection accuracy metrics.

This is precisely the kind of failure that a standard evaluation pipeline will not catch: Precision, Recall, and F1-score describe whether obstacles were detected, not whether the safety threshold applied to them was correct for the deployment hardware. A system could score well on all three metrics while still misjudging distance badly enough to alert too late, or too often, once deployed on unfamiliar hardware. We view this gap, between what standard metrics certify and what the system actually guarantees to the end user, as the central safety-relevant finding of this project.

We documented this limitation rather than silently patching it, and began investigating per-camera calibration procedures (e.g., a short calibration step using a known reference distance at setup time) to make the depth-to-distance conversion robust across devices before any real-world deployment.

5. Discussion and Future Work

Vision-Mate demonstrates that fusing detection and depth models is feasible for real-time assistive alerting, but also illustrates a broader pattern: safety-critical assumptions (here, a calibration constant) can be buried inside a pipeline stage that looks, from the outside, like a solved engineering problem. Future work includes an automatic or user-guided calibration routine, evaluation of distance-estimation error (not just detection accuracy) across multiple camera models, and stress-testing the alert system under degraded conditions such as low light or partial occlusion.

6. Conclusion

Vision-Mate is a working real-time obstacle-alert system, but its more important contribution here is diagnostic: it surfaces how a single unexamined assumption in a depth-estimation pipeline can undermine the safety guarantee a system appears to provide, without showing up in conventional accuracy metrics. That distinction, between a model that performs well and a system that is safe to rely on, is the throughline this project left me most interested in pursuing further.